SWING: Source-Reweighting-Induced Eigen-Shrinkage for Single-Run Post-hoc Domain Generalization

Anonymous Authors

Abstract

Single-run post-hoc domain generalization is attractive because it can be attached to a strong empirical-risk-minimization pipeline without retraining. Existing post-hoc methods in this regime predominantly select or average checkpoints inside one connected low-loss region, motivated by flatness, locality, or diversity. This paper studies a different remaining degree of freedom: *once a safe basin has already been found, should all directions inside that basin be trusted equally under domain shift?* We propose **SWING** (**S**ource-re**W**eighting-**I**nduced ei**N**-shrin**G**age), a single-run, post-hoc, domain-label-free method that learns a low-rank basin geometry from one dense checkpoint trajectory and then contracts only the directions that are fragile under plausible source-support reweightings. The method first constructs a connected checkpoint pool, defines a basin center and trajectory subspace, fits a local quadratic source-risk surrogate on one held-out split, estimates a reweighting fragility operator on a disjoint split, and finally outputs a single model by minimizing the resulting surrogate inside a basin trust region; when the constraint is inactive, this reduces to generalized eigen-shrinkage of the source-fit solution. We prove a local hidden-shift upper bound in which the reweighting term appears as a square-root variance penalty, derive the quadratic SWING surrogate from that bound, show that its unconstrained minimizer has an exact generalized eigen-shrinkage form, give a precise comparison with scalar shrinkage inside the quadratic model, and establish a deterministic perturbation bound together with a conditional plug-in consistency statement for the unconstrained estimator. The experimental sections intentionally leave all new SWING quantitative results as **TBD**, while recording exact published benchmark context from prior work and providing the full evaluation protocol.

1 Introduction

Domain generalization (DG) asks for a predictor that performs well on an unseen target environment using only source data at training time. In practice, the DG methods that last are often the ones that preserve the base training recipe and change only how the final model is selected. This is why single-run post-hoc selection matters: one can train once, save dense checkpoints, and choose a deployable model after training has already finished.

This regime already has a clear reference point. On the standard DomainBed ResNet-50 benchmark suite, SWAD improves the average out-of-domain accuracy of ERM from 63.3 to 66.9 across PACS, VLCS, OfficeHome, TerraIncognita, and DomainNet [3]. More diverse weight-averaging schemes can go further in heavier settings, with DiWA reporting 68.0 in its strongest multi-run, multi-model configuration [18], and a pretraining-aware method combined with the same post-hoc selector reaching 68.1 [4]. The empirical lesson is clear: the final predictor is often not the last checkpoint.

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Yet existing post-hoc DG methods still work inside one narrow action family. They select a checkpoint, an interval, or a soup inside a connected low-loss region. That is often enough to find a safe basin, but it leaves a natural question unanswered:

> **After a safe basin has been identified, should the final model keep every direction inside that basin at full strength?**

Our answer is no. Some directions inside a good basin are strongly supported by the pooled source data; others are fragile because small source-support redistributions would swing their contribution substantially. Those fragile directions are exactly the ones that should be attenuated before deployment.

This principle leads to **SWING** (**S**ource-re**W**eighting-**I**nduced ei**N**-shrin**G**age). SWING has two stages. First, it builds a safe connected checkpoint pool, computes a basin center, and extracts a low-rank trajectory subspace. Second, inside that subspace, it fits:

- a *source-fit operator* A that captures how strongly the pooled source data supports each direction, and
- a *fragility operator* B that measures how much examplewise local loss slopes vary across the source support, hence how strongly nearby source reweightings can swing the direction.

The final model is a single deterministic weight vector obtained by minimizing the surrogate inside the basin trust region. When the trust-region constraint is inactive, the edit reduces to shrinking the fit solution along the generalized eigenvectors of the pair (B, A).

The central idea may be stated in one sentence:

> **Directions that swing under plausible source reweightings should not survive deployment at full strength.**

This move is still grounded in the current post-hoc DG literature. Weight averaging, mode connectivity, and soups tell us that dense trajectories often lie in a common low-loss basin [8, 9, 12, 21]. Subspace-based Bayesian approximations further show that a small trajectory subspace can contain many high-performing models [13]. Influence analyses and distributionally robust optimization show that sample reweightings are a natural first-order model of nearby distribution shifts [5, 7, 15]. SWING combines these threads into a post-hoc DG method that is still simple, still single-model at test time, and more directly tied to hidden shift than isotropic averaging.

We make five contributions.

1. We introduce SWING, a single-run, post-hoc, domain-label-free DG method that performs a full-weight basin edit rather than another checkpoint selector.
2. We derive SWING from a local hidden-shift model in which target domains are source-supported χ^2-bounded reweightings of the pooled source support. This yields a square-root fragility term $\sqrt{\varrho\, z^\top B z}$ and, after quadratic majorization, the SWING surrogate.
3. We prove that the unconstrained SWING edit has an exact generalized eigen-shrinkage form, and that the deployed estimator inherits this form whenever the trust-region constraint is inactive. We also give a precise within-model comparison against scalar shrinkage.
4. We prove a deterministic perturbation bound and a conditional plug-in consistency statement for the unconstrained empirical estimator, and give a restricted-family target-risk certificate for the returned basin edit.
5. We provide a complete NeurIPS-style manuscript with original analytical figures, a reproducible protocol, and published baseline context from prior work; all new SWING quantitative results remain explicitly marked **TBD**.

2 Related Work

Single-run post-hoc DG. SWAD is the canonical single-run post-hoc DG method [3]. Its practical appeal comes from dense checkpointing, overfit-aware windowing, and a single averaged model

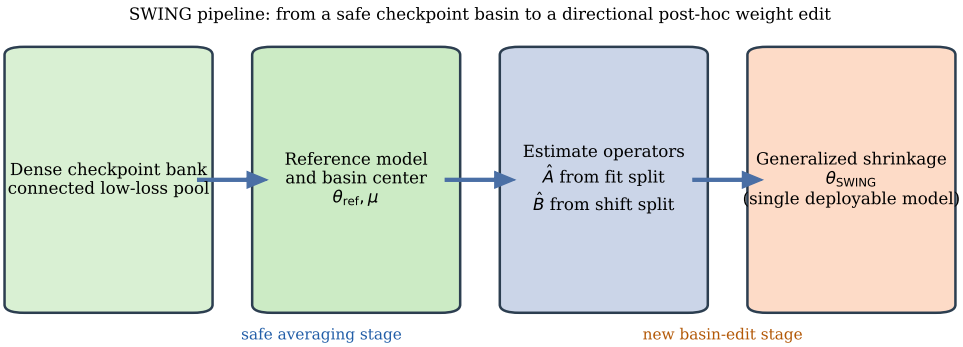


Figure 1: **SWING pipeline.** A dense checkpoint bank is reduced to a connected safe pool. That pool defines a basin center μ and a low-rank subspace U. On disjoint source splits, SWING estimates a source-fit operator $\hat{A}$ and a reweighting fragility operator $\hat{B}$, then outputs one deployable model by minimizing the surrogate within the basin trust region; when the constraint is inactive, this reduces to generalized eigen-shrinkage.

at test time. Published follow-ups reinforce that post-hoc averaging is genuinely strong. At the same time, the strongest published DIWA configuration is explicitly heavier, averaging models from different training runs [18], while MIRO improves the training representation and then benefits again from post-hoc selection [4]. SWING stays in the single-run deployment regime and changes the post-hoc action itself: it edits the final weight vector inside a safe basin rather than merely choosing which checkpoints to average.

Weight averaging, connectivity, and basin geometry. Stochastic weight averaging [12], fast geometric ensembling [9], linear mode connectivity [8], and model soups [21] show that dense training trajectories often traverse connected low-loss regions. Subspace inference further shows that a small trajectory subspace can contain many strong models [13]. SWING uses these ideas as infrastructure: a connected checkpoint pool defines the basin, and a low-rank subspace provides the coordinate system for post-hoc editing.

Reweighting, influence, and local robustness. Influence functions quantify local sensitivity to sample reweightings [15]. Distributionally robust optimization studies worst-case risk over ambiguity sets such as χ^2 or other f-divergence balls [5–7]. Closely related literatures study worst-case subpopulations, fairness without observed demographics, and group-robust learning through reweighting [11, 16, 17, 19]. Tangent Prop is the classical precursor in input space: useful invariance can be enforced by controlling derivatives along nuisance directions [20]. SWING transfers this logic to the empirical distribution: its nuisance directions are nearby redistributions of the source support.

Adjacent post-training weight editing outside DG. There is recent interest in editing weights after training to improve robustness or preserve OOD performance, including WiSE-FT interpolation, robustness signatures, model-stock averaging, and spectral soup editing [1, 2, 14, 22]. These works are adjacent but different. They do not target single-run post-hoc DG with source-only held-out data, do not derive a hidden-subpopulation reweighting geometry, and do not formulate the final edit as the solution of a basin-restricted hidden-shift surrogate whose unconstrained form is generalized eigen-shrinkage.

3 Problem Setup

We work in the standard source-only DG setting. Let P denote the pooled source distribution over examples $Z = (X, Y)$. A dense training run produces checkpoints $\theta_1, \ldots, \theta_T \in \mathbb{R}^d$. The method is post-hoc: training is already over, and only the checkpoints and held-out source data are used.

Safe checkpoint pool. Let S_{pool} be a held-out source split used for pool construction. Define a source validation loss $\widehat{L}_{\text{pool}}(\theta)$ on this split. Let

$$t^\star \in \arg\min_{1\le t\le T} \widehat{L}_{\text{pool}}(\theta_t)$$

be the best checkpoint. For checkpoints i, j, define the empirical interpolation barrier

$$\text{bar}(i,j) = \max_{\alpha\in\mathcal{G}} \widehat{L}_{\text{pool}}\left(\alpha\theta_i + (1-\alpha)\theta_j\right) - \max\left\{\widehat{L}_{\text{pool}}(\theta_i), \widehat{L}_{\text{pool}}(\theta_j)\right\},$$

where $\mathcal{G} \subset [0,1]$ is a fixed interpolation grid. For user-chosen thresholds $\varepsilon_{\text{pool}}, \tau_{\text{pool}} > 0$, define the safe pool

$$\mathcal{C} = \left\{t : \widehat{L}_{\text{pool}}(\theta_t) \le \widehat{L}_{\text{pool}}(\theta_{t^\star}) + \varepsilon_{\text{pool}},\ \text{bar}(t, t^\star) \le \tau_{\text{pool}}\right\}.$$

This pool is intended to isolate one connected low-loss basin. The pool construction stage is not the novelty of the paper; it is the infrastructure on top of which SWING operates.

Basin coordinates. Define the basin center

$$\mu = \frac{1}{|\mathcal{C}|}\sum_{t\in\mathcal{C}} \theta_t.$$

Let $U \in \mathbb{R}^{d\times r}$ have orthonormal columns spanning the top r principal components of $\{\theta_t - \mu : t \in \mathcal{C}\}$. Every point $z \in \mathbb{R}^r$ induces a full model

$$\theta(z) = \mu + Uz.$$

The safe pool itself induces observed coordinates

$$z_t = U^\top(\theta_t - \mu), \qquad t \in \mathcal{C}.$$

Disjoint source splits. Beyond S_{pool}, SWING uses two disjoint held-out source splits:

- S_{fit}, used to fit a local quadratic source-risk surrogate;
- S_{shift}, used to estimate the fragility operator.

The split separation is deliberate: it prevents the same source examples from simultaneously defining both the fit geometry and the hidden-shift geometry.

In theory, $\mathcal{Z}$ is any compact trust region containing the observed safe-pool coordinates. In practice, we instantiate $\mathcal{Z}$ as a box in generalized coordinates whose sides are determined by the extrema of the safe-pool projections.

4 The Method

4.1 Quadratic source-fit surrogate

On S_{fit}, SWING fits a local quadratic model of source risk in the basin subspace:

$$q(z) = c + a^\top z + \frac{1}{2} z^\top A z, \qquad A \in \mathbb{R}^{r\times r},\ A \succ 0.$$

In practice, q is estimated from the observed checkpoint cloud $\{z_t : t \in \mathcal{C}\}$ by least-squares fitting of the empirical losses

$$\widehat{L}_{\text{fit}}(\theta_t) \approx c + a^\top z_t + \frac{1}{2} z_t^\top A z_t.$$

The unique minimizer of the quadratic fit-only surrogate is

$$z_{\text{fit}} = -A^{-1}a.$$

This is the source-fit solution in the basin coordinate system.

4.2 Fragility operator from source reweightings

Fix an example $Z_i \in S_{\text{shift}}$. Let $\phi_i \in \mathbb{R}^r$ denote the local first-order loss slope of that example in the basin subspace. Formally, in the theory we assume

$$\ell\left(\theta(z); Z_i\right) = \ell\left(\theta(0); Z_i\right) + \phi_i^\top z + r_i(z),$$

with a controlled quadratic remainder $r_i(z)$.

The fragility operator is the covariance of these examplewise slopes:

$$B = \mathrm{Cov}(\phi) = \mathbb{E}\left[(\phi - \mathbb{E}[\phi])(\phi - \mathbb{E}[\phi])^\top\right].$$

Intuitively:

- if a direction has small variance in examplewise slopes, then nearby source reweightings do not change its preferred update much;
- if a direction has large variance, then small hidden redistributions of the source support can swing its utility sharply.

In practice, ϕ_i is estimated from the checkpoint cloud by regressing the examplewise losses of Z_i against the coordinates z_t, $t \in \mathcal{C}$, and $\hat{B}$ is the sample covariance of the fitted slope vectors.

4.3 The SWING objective

The derivation in Section 5 yields the quadratic robust surrogate

$$J_\lambda(z) = a^\top z + \frac{1}{2} z^\top A z + \frac{\lambda}{2} z^\top B z, \qquad \lambda > 0.$$

The first two terms retain source fit. The third term penalizes directions whose first-order contribution is unstable under plausible source-support reweightings.

The theoretical returned estimator is the constrained minimizer

$$z_\lambda^{\mathcal{Z}} \in \arg\min_{z \in \mathcal{Z}} J_\lambda(z).$$

Whenever the unconstrained minimizer lies inside $\mathcal{Z}$, the constrained and unconstrained solutions coincide.

Let $z_{\text{fit}} = -A^{-1}a$. Then

$$J_\lambda(z) = \text{const} + \frac{1}{2}(z - z_{\text{fit}})^\top A (z - z_{\text{fit}}) + \frac{\lambda}{2} z^\top B z.$$

Therefore SWING should be interpreted as a *basin edit*: it does not search for a new basin, but contracts the fit solution along fragile directions inside a basin that was already judged safe by the first-stage pool construction.

4.4 Generalized eigen-shrinkage

Because $A \succ 0$ and $B \succeq 0$, there exists an A-orthonormal generalized eigenbasis $V = [v_1, \ldots, v_r]$ and nonnegative generalized eigenvalues $\nu_1, \ldots, \nu_r$ such that

$$B v_j = \nu_j A v_j, \qquad V^\top A V = I.$$

Writing

$$z_{\text{fit}} = \sum_{j=1}^{r} c_j v_j, \qquad c_j = v_j^\top A z_{\text{fit}},$$

the unconstrained SWING edit is

$$z_\lambda^{\text{unc}} = \sum_{j=1}^{r} \frac{c_j}{1 + \lambda \nu_j} v_j.$$

Thus every generalized coordinate is shrunk by a factor

$$\beta_j(\lambda) = \frac{1}{1 + \lambda \nu_j}.$$

Directions with larger fragility-to-fit ratio ν_j receive stronger attenuation.

Algorithm 1 SWING

Require: Dense checkpoint bank $\{\theta_t\}_{t=1}^{T}$, held-out source splits $S_{\text{pool}}, S_{\text{fit}}, S_{\text{shift}}$, rank r, thresholds $\varepsilon_{\text{pool}}, \tau_{\text{pool}}$, regularization λ.

Select the safe checkpoint pool $\mathcal{C}$ using source-fit thresholding and interpolation barriers on S_{pool}.
Compute the basin center $\mu = |\mathcal{C}|^{-1} \sum_{t \in \mathcal{C}} \theta_t$.
Compute $U \in \mathbb{R}^{d \times r}$ from the top r principal directions of $\{\theta_t - \mu : t \in \mathcal{C}\}$.
Form checkpoint coordinates $z_t = U^\top(\theta_t - \mu)$, $t \in \mathcal{C}$.
Fit the quadratic source-risk surrogate $q(z) = c + a^\top z + \frac{1}{2} z^\top A z$ on S_{fit}.
Fit examplewise slope vectors $\hat{\phi}_i$ on S_{shift} and set $\hat{B} = \widehat{\text{Cov}}(\hat{\phi})$.
Solve the generalized eigenproblem $\hat{B} v_j = \nu_j \hat{A} v_j$, $V^\top \hat{A} V = I$.
Compute $z_{\text{fit}} = -\hat{A}^{-1}\hat{a}$ and coefficients $c_j = v_j^\top \hat{A} z_{\text{fit}}$.
Set the unconstrained edit $z_\lambda^{\text{unc}} = \sum_{j=1}^{r} \frac{c_j}{1+\lambda\nu_j} v_j$.
If $z_\lambda^{\text{unc}} \in \mathcal{Z}$, set $z_\lambda^{\mathcal{Z}} = z_\lambda^{\text{unc}}$; otherwise solve $z_\lambda^{\mathcal{Z}} \in \arg\min_{z \in \mathcal{Z}} J_\lambda(z)$ inside the chosen trust region.
return $\theta_{\text{SWING}} = \mu + U z_\lambda^{\mathcal{Z}}$.

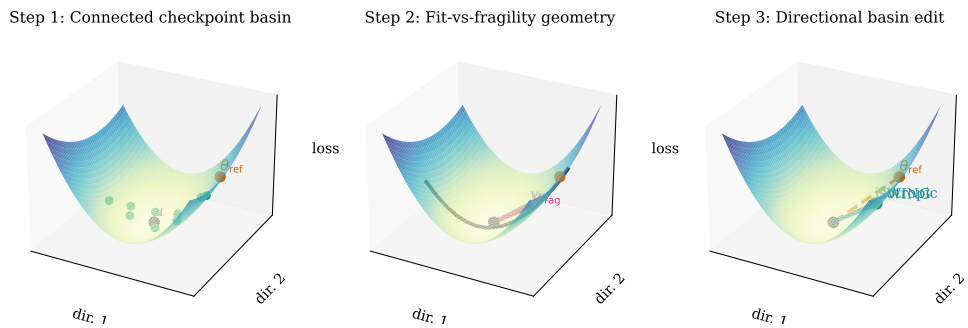


Figure 2: **Three-dimensional intuition for SWING.** Left: dense checkpoints define a connected low-loss basin and a center μ. Middle: the basin subspace supplies the coordinates in which fit and fragility are estimated. Right: SWING contracts the fit solution anisotropically, suppressing high-fragility directions more than low-fragility directions.

5 Theory

This section derives the SWING objective from a local hidden-shift model and proves its main properties. The assumptions are deliberately explicit. The theory is local and subspace-restricted; it does not claim that arbitrary deep networks are globally quadratic in weight space.

5.1 Assumptions

Assumption 1 (Local quadratic source-fit model)**.** *There exists a compact set $\mathcal{Z} \subset \mathbb{R}^r$ containing the observed checkpoint coordinates and a positive definite matrix $A \in \mathbb{R}^{r \times r}$ and vector $a \in \mathbb{R}^r$ such that the pooled source excess risk on the basin subspace satisfies*

$$L_P(z) - L_P(0) = a^\top z + \frac{1}{2} z^\top A z \qquad \textit{for all } z \in \mathcal{Z}.$$

Assumption 2 (Examplewise local linearization with quadratic remainder)**.** *There exist measurable $\phi : \mathsf{Z} \to \mathbb{R}^r$ and $\kappa : \mathsf{Z} \to [0, \infty)$ such that for every example $Z \in \mathsf{Z}$ and every $z \in \mathcal{Z}$,*

$$\ell(\theta(z); Z) = \ell(\theta(0); Z) + \phi(Z)^\top z + r(Z, z),$$

where the remainder satisfies

$$|r(Z, z)| \le \frac{\kappa(Z)}{2} \|z\|^2 \qquad \textit{for all } z \in \mathcal{Z}.$$

Moreover $\mathbb{E}_P[\|\phi(Z)\|^2] < \infty$ and $\mathbb{E}_P[\kappa(Z)^2] \le K^2 < \infty$.

Assumption 3 (Source-supported local target shift)**.** *The target distribution Q is absolutely continuous with respect to P and belongs to the local χ^2-style ambiguity family*

$$\mathcal{Q}(\varrho) = \left\{ Q : \frac{dQ}{dP} = 1 + h,\ 1 + h \geq 0\ P\text{-a.s.},\ \mathbb{E}_P[h] = 0,\ \mathbb{E}_P[h^2] \leq \varrho \right\}$$

for some radius $\varrho \geq 0$.

Assumption 3 encodes the source-supported hidden-subpopulation view: unseen domains are not arbitrary, but are local redistributions of source support mass. Assumption 2 makes the reweighting effect observable through the examplewise slope covariance. Assumption 1 isolates the source-fit geometry inside a single connected basin.

5.2 Deriving the fragility term

Theorem 1 (Local hidden-shift upper bound)**.** *Under Assumptions 1–3, define*

$$\bar{\phi} = \mathbb{E}_P[\phi(Z)], \qquad B = \mathrm{Cov}_P(\phi(Z)) = \mathbb{E}_P\left[(\phi(Z) - \bar{\phi})(\phi(Z) - \bar{\phi})^\top\right].$$

Then for every $z \in \mathcal{Z}$ and every $Q \in \mathcal{Q}(\varrho)$,

$$L_Q(z) \leq L_Q(0) + a^\top z + \frac{1}{2} z^\top A z + \sqrt{\varrho\, z^\top B z} + \frac{\sqrt{\varrho}\, K}{2} \|z\|^2.$$

The theorem shows where the fragility term comes from. The quantity $z^\top B z$ is the variance of the first-order examplewise slope in direction z. If that variance is large, local source reweightings can swing the contribution of z strongly; if it is small, the direction is reweighting-stable.

Corollary 1 (Quadratic SWING surrogate)**.** *Under the assumptions of Theorem 1, for every $\lambda > 0$,*

$$L_Q(z) \leq L_Q(0) + \frac{\varrho}{2\lambda} + a^\top z + \frac{1}{2} z^\top \widetilde{A} z + \frac{\lambda}{2} z^\top B z \qquad \textit{for all } z \in \mathcal{Z},$$

where

$$\widetilde{A} = A + \sqrt{\varrho}\, K I_r.$$

Corollary 1 is the exact bridge from the hidden-shift model to SWING: the practical objective is the smallest quadratic upper envelope obtained by balancing the source-fit term and the reweighting fragility term.

5.3 Closed-form shrinkage solution

For the rest of the theory, write A for the positive definite matrix $\widetilde{A}$ from Corollary 1; this only lightens notation.

Theorem 2 (Generalized eigen-shrinkage)**.** *Let*

$$J_\lambda(z) = a^\top z + \frac{1}{2} z^\top A z + \frac{\lambda}{2} z^\top B z, \qquad A \succ 0,\ B \succeq 0,\ \lambda > 0.$$

Define the fit-only solution

$$z_{\mathrm{fit}} = -A^{-1} a.$$

Then:

1. *The unconstrained problem $\min_{z \in \mathbb{R}^r} J_\lambda(z)$ is strongly convex and has the unique minimizer*
$$z_\lambda^{\mathrm{unc}} = -(A + \lambda B)^{-1} a = (A + \lambda B)^{-1} A z_{\mathrm{fit}}.$$
2. *If $z_\lambda^{\mathrm{unc}} \in \mathcal{Z}$, then $z_\lambda^{\mathcal{Z}} = z_\lambda^{\mathrm{unc}}$. In that case, if $V = [v_1, \ldots, v_r]$ is an A-orthonormal generalized eigenbasis satisfying*
$$B v_j = \nu_j A v_j, \qquad V^\top A V = I, \qquad \nu_j \geq 0,$$
and
$$z_{\mathrm{fit}} = \sum_{j=1}^{r} c_j v_j, \qquad c_j = v_j^\top A z_{\mathrm{fit}},$$
then
$$z_\lambda^{\mathcal{Z}} = z_\lambda^{\mathrm{unc}} = \sum_{j=1}^{r} \frac{c_j}{1 + \lambda \nu_j} v_j.$$

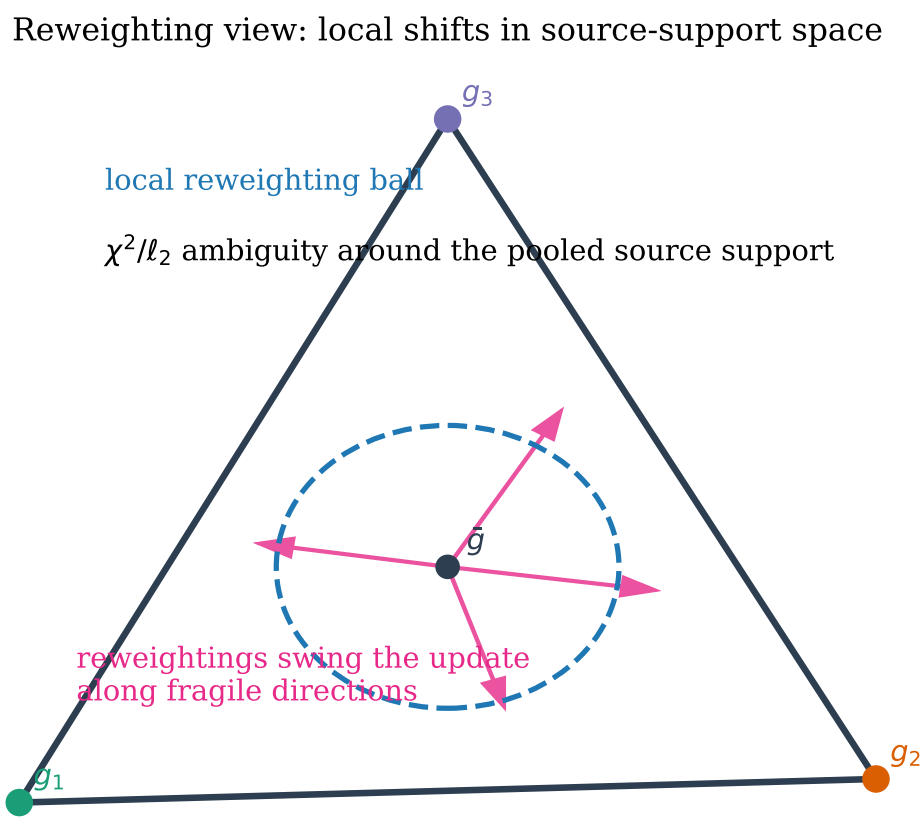


Figure 3: **Why reweighting induces a fragility operator.** Nearby target domains are modeled as local redistributions of source-support mass. Examplewise slope vectors that fan out strongly under these redistributions produce a large covariance B, so directions aligned with them are shrunk more aggressively by SWING.

Theorem 2 is the central mathematical result of the paper. It shows that SWING is not merely a heuristic penalty. Whenever the unconstrained minimizer stays inside the chosen basin trust region, the returned solution is an exact contraction of the fit-only solution, with one shrinkage factor per generalized fragility level.

5.4 Why anisotropy matters

The next comparison result formalizes an important point. If generalized fragility levels are heterogeneous, then one global scalar shrinkage factor is not enough inside the quadratic model.

Proposition 1 (Comparison to scalar shrinkage)**.** *Under the assumptions of Theorem 2, consider the scalar shrinkage family*

$$\mathcal{F}_{\mathrm{sc}} = \{\alpha z_{\mathrm{fit}} : \alpha \in \mathbb{R}\}.$$

Then

$$\min_{\alpha \in \mathbb{R}} J_\lambda(\alpha z_{\mathrm{fit}}) \geq J_\lambda(z_\lambda^{\mathrm{unc}}).$$

Moreover equality holds if and only if the generalized eigenvalues ν_j are constant on the support of z_{fit}, that is,

$$\nu_j = \nu_k \qquad \text{for all } j, k \text{ such that } c_j \neq 0 \text{ and } c_k \neq 0.$$

Consequently, whenever z_{fit} has nonzero mass on at least two distinct generalized fragility levels, anisotropic shrinkage is strictly better than the best scalar shrinkage.

Proposition 1 records the simplest exact comparison that motivates anisotropy in the quadratic surrogate. Scalar shrinkage is optimal only in the degenerate case where all active directions have the same fragility-to-fit ratio.

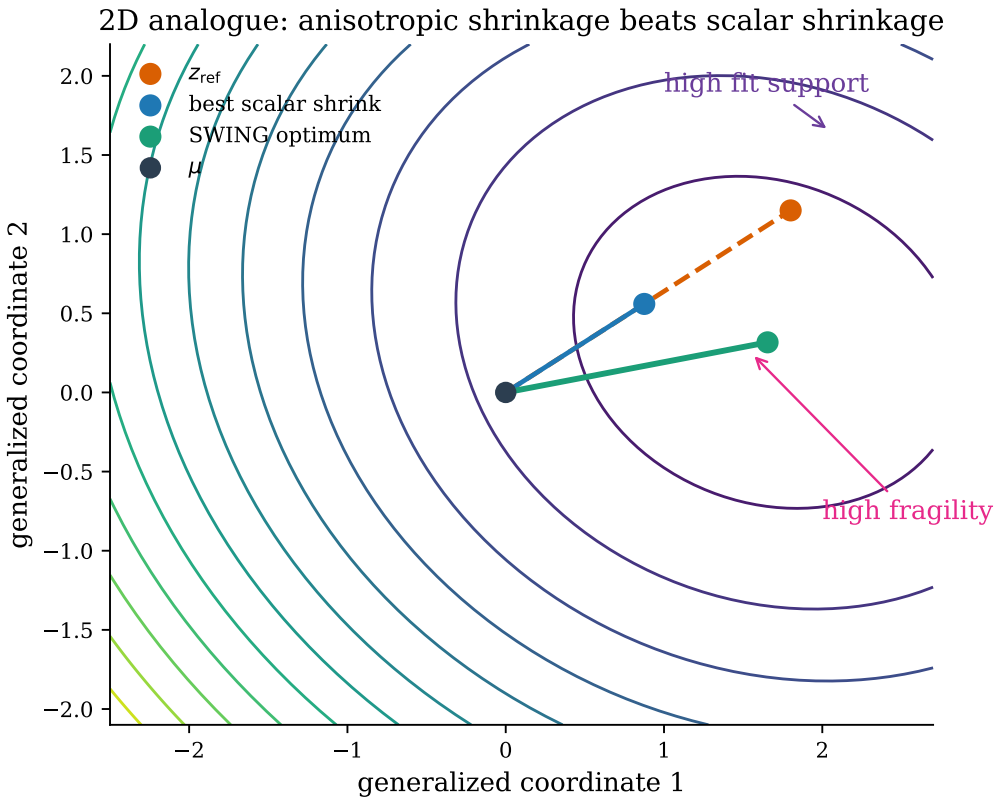


Figure 4: **Two-dimensional analogue of Proposition 1.** The fit solution z_{fit} lies away from the basin center μ. The best scalar shrinkage moves only along the dashed ray. The SWING optimum follows the anisotropic geometry defined by A and B, producing a lower robust surrogate value whenever fragility is direction-dependent.

5.5 Certificate improvement and conditional stability

Proposition 2 (Exact certificate gain over the fit-only solution)**.** *Under the assumptions of Theorem 2,*

$$J_\lambda(z_{\text{fit}}) - J_\lambda(z_\lambda^{\text{unc}}) = \frac{1}{2}(z_{\text{fit}} - z_\lambda^{\text{unc}})^\top (A + \lambda B)(z_{\text{fit}} - z_\lambda^{\text{unc}}).$$

In particular,

$$J_\lambda(z_\lambda^{\text{unc}}) \le J_\lambda(z_{\text{fit}}),$$

with equality if and only if $Bz_{\text{fit}} = 0$.

Proposition 2 quantifies the exact improvement that SWING certifies over the fit-only solution. If the fit solution already lies in the kernel of the fragility operator, there is nothing to shrink; otherwise the robust surrogate is improved strictly.

Theorem 3 (Deterministic perturbation bound)**.** *Let*

$$M = A + \lambda B, \qquad \hat{M} = \hat{A} + \lambda \hat{B}, \qquad m = \lambda_{\min}(M).$$

Suppose $m > 0$ *and define*

$$\delta = \left\|\hat{A} - A\right\| + \lambda \left\|\hat{B} - B\right\|.$$

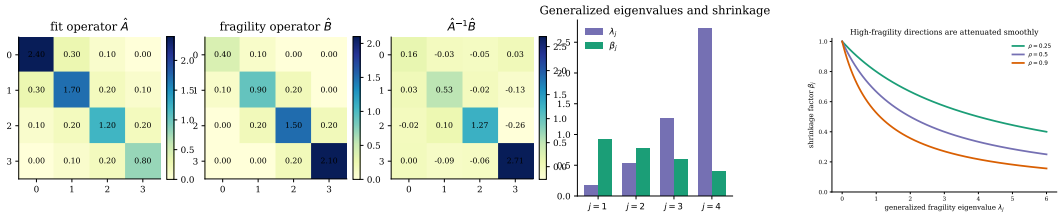
If $\delta < m$*, then* $\hat{M}$ *is positive definite and the empirical solution*

$$\hat{z}_\lambda = -\hat{M}^{-1}\hat{a}$$

is well defined. Moreover

$$\|\hat{z}_\lambda - z_\lambda^{\text{unc}}\| \le \frac{\|\hat{a} - a\| + \delta \|z_\lambda^{\text{unc}}\|}{m - \delta}.$$

Consequently, if $\hat{a} \to a$*,* $\hat{A} \to A$*, and* $\hat{B} \to B$ *in operator norm, then* $\hat{z}_\lambda \to z_\lambda^{\text{unc}}$.



(a) Fit and fragility operators, their generalized geometry, and the resulting spectrum.

(b) Shrinkage path as fragility rises.

Figure 5: **Operator-level intuition for SWING.** The fit operator $\hat{A}$ says which directions the source data trusts. The fragility operator $\hat{B}$ says which directions swing under source-support redistributions. Their generalized eigensystem defines the contraction geometry.

Theorem 3 is the paper's main convergence statement. It is a deterministic continuity bound: conditional on consistent operator estimators, the resulting unconstrained SWING estimator is also consistent, and this directly covers the deployed edit whenever the trust-region constraint is inactive.

Corollary 2 (Restricted-family target certificate)**.** *Under Assumptions 1–3, let*

$$z_\lambda^{\mathcal{Z}} \in \arg\min_{z\in\mathcal{Z}} J_\lambda(z).$$

Then for every $Q \in \mathcal{Q}(\varrho)$ and every $z \in \mathcal{Z}$,

$$L_Q(z_\lambda^{\mathcal{Z}}) \le L_Q(0) + \frac{\varrho}{2\lambda} + J_\lambda(z_\lambda^{\mathcal{Z}}) \le L_Q(0) + \frac{\varrho}{2\lambda} + J_\lambda(z).$$

In particular, the SWING *solution attains the smallest available quadratic hidden-shift certificate inside the chosen basin family.*

6 Practical Estimation Details

Estimating the quadratic fit surrogate. Given safe-pool coordinates z_t and fit-split losses $\widehat{L}_{\text{fit}}(\theta_t)$, we estimate $(\hat{c}, \hat{a}, \hat{A})$ by solving

$$\min_{c,a,A=A^\top} \sum_{t\in\mathcal{C}} \left(\widehat{L}_{\text{fit}}(\theta_t) - c - a^\top z_t - \frac{1}{2} z_t^\top A z_t \right)^2 + \eta_A \|A\|_F^2$$

subject to $\hat{A} \succeq m_A I_r$ for a small ridge floor $m_A > 0$. This makes the source-fit surrogate strictly convex and numerically stable.

Estimating the slope covariance. For each example $Z_i \in S_{\text{shift}}$, let $\ell_{it} = \ell(\theta_t; Z_i)$. We estimate the examplewise slope vector $\hat{\phi}_i$ from the linear model

$$\ell_{it} - \bar{\ell}_i \approx \hat{\phi}_i^\top z_t, \qquad \bar{\ell}_i = \frac{1}{|\mathcal{C}|} \sum_{t\in\mathcal{C}} \ell_{it},$$

using ridge regression over $t \in \mathcal{C}$. Then

$$\hat{B} = \frac{1}{|S_{\text{shift}}|} \sum_{i\in S_{\text{shift}}} (\hat{\phi}_i - \bar{\phi})(\hat{\phi}_i - \bar{\phi})^\top, \qquad \bar{\phi} = \frac{1}{|S_{\text{shift}}|} \sum_i \hat{\phi}_i.$$

Computational complexity. If d is the ambient parameter dimension, $r \ll d$ is the subspace rank, $M = |\mathcal{C}|$ is the safe-pool size, and $n_s = |S_{\text{shift}}|$, then the dominant costs are:

- subspace extraction from M checkpoints;
- n_s ridge regressions in r variables to estimate slopes;
- one $r \times r$ generalized eigendecomposition.

The final deployment cost remains unchanged from ordinary single-model inference because SWING outputs one weight vector.

Method	PACS	VLCS	OfficeHome	TerraInc.	DomainNet	Avg.
ERM [3]	85.5	77.5	66.5	46.1	40.9	63.3
SWAD [3]	88.1	79.1	70.6	50.0	46.5	66.9
DiWA† [18]	89.0	78.6	72.8	51.9	47.7	68.0
MIRO + SWAD [4]	88.4	79.6	72.4	52.9	47.0	68.1

Table 1: Published benchmark context from prior work. † DiWA's strongest published configuration uses a heavier multi-model setting than pure single-run post-hoc selection.

Method	PACS	VLCS	OfficeHome	TerraInc.	DomainNet	Avg.
ERM	85.5	77.5	66.5	46.1	40.9	63.3
SWAD	88.1	79.1	70.6	50.0	46.5	66.9
SWING	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**

Table 2: Planned primary comparison for the pure single-run post-hoc setting. All SWING entries are intentionally left as **TBD** until experiments are run.

7 Experimental Protocol

All new empirical results for SWING are intentionally left as **TBD**. This section records the protocol, published benchmark context, and the exact ablations the eventual empirical study should run.

7.1 Benchmarks and training regime

We will evaluate on the standard DomainBed suite [10]:

- PACS,
- VLCS,
- OfficeHome,
- TerraIncognita,
- DomainNet.

The intended baseline regime is the standard ResNet-50 single-run post-hoc setting established by published SWAD comparisons. The DiWA row is retained only as broader published context from a heavier setting, not as the pure single-run reference regime. The checkpoint bank should be dense enough to resolve one late connected basin. All post-hoc selection and shrinkage must use only source-held-out data and must not use source-domain identities.

7.2 Published benchmark context

Table 1 records exact published numbers from prior work. These are not new SWING results; they define the bar that any serious single-run post-hoc DG method must eventually clear.

7.3 Planned main table

7.4 Required ablations

The empirical study should include at least the following ablations.

1. **No fragility term:** set $B = 0$ and recover the pure fit solution.
2. **Scalar shrinkage:** replace generalized shrinkage with the best one-parameter interpolation toward μ.
3. **No safe pool:** build U from a broader late trajectory without the barrier screen.
4. **Rank sensitivity:** vary r.
5. **Regularization sensitivity:** vary λ.
6. **Split overlap test:** compare disjoint versus reused fit/shift splits.

Variant	Avg. Acc.
SWING	**TBD**
No fragility term ($B = 0$)	**TBD**
Best scalar shrinkage	**TBD**
No safe pool screen	**TBD**
Small rank r	**TBD**
Large rank r	**TBD**
Small λ	**TBD**
Large λ	**TBD**

Table 3: Planned ablation table. All new numbers remain **TBD**.

8 Discussion

What SWING is and is not. SWING is not another global soup optimizer. It is also not a last-layer repair method, not a multi-run averaging scheme, and not a target-time adaptation method. It is a full-weight basin edit that acts only after a safe post-hoc basin has already been found.

Why the theory is local. The paper's theory is intentionally restricted to a connected low-rank basin family. That is a strength, not a weakness. The whole operational point of post-hoc single-run DG is that one dense trajectory often already contains a safe region; the scientifically honest question is what can be proved *inside that region*.

Limitations. The theory assumes a source-supported local shift model. If the target domain introduces genuinely new support, then source-reweighting fragility cannot fully characterize deployment risk. The quadratic source-fit model is also local, and the regression estimators for A and B can become noisy if the safe pool is too small or too narrow. Finally, the eventual empirical study must verify that the gains are not captured by simpler scalar shrinkage.

9 Conclusion

SWING is a simple claim made operational: once a good checkpoint basin has been found, not all directions inside that basin should be trusted equally. Directions whose contribution swings under plausible source-support reweightings should be shrunk. The resulting method stays single-run, post-hoc, domain-label-free, and single-model at test time. Its math is explicit: a hidden-shift upper bound yields a square-root fragility term, a quadratic surrogate, the exact generalized eigen-shrinkage form of the unconstrained basin edit, a clean comparison against scalar shrinkage inside the quadratic model, and a deterministic perturbation theory that upgrades to consistency for the unconstrained estimator under consistent operator estimation. The empirical study remains to be run; the paper, figures, proofs, and protocol are ready.

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A Additional Geometric Intuition

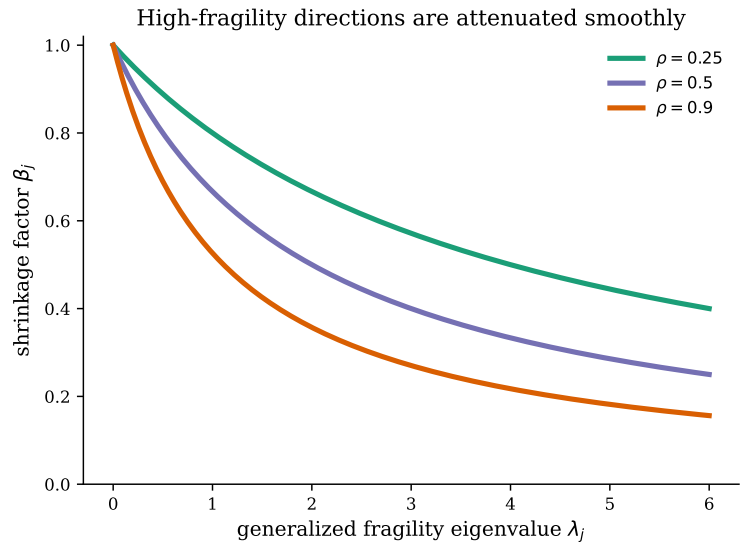

Figure 6: **Regularization path.** Each generalized direction is shrunk by $\beta_j(\lambda) = 1/(1 + \lambda\nu_j)$. Larger fragility eigenvalues decay faster as λ grows.

B Proofs

B.1 Proof of Theorem 1

Proof. Fix $z \in \mathcal{Z}$ and $Q \in \mathcal{Q}(\varrho)$. By Assumption 3, there exists h such that $dQ/dP = 1 + h$, $\mathbb{E}_P[h] = 0$, and $\mathbb{E}_P[h^2] \leq \varrho$. Therefore

$$\begin{aligned} L_Q(z) - L_Q(0) &= \mathbb{E}_Q\left[\ell(\theta(z); Z) - \ell(\theta(0); Z)\right] \\ &= \mathbb{E}_P\left[(1 + h(Z))\left(\ell(\theta(z); Z) - \ell(\theta(0); Z)\right)\right]. \end{aligned} \tag{1}$$

By Assumption 2,

$$\ell(\theta(z); Z) = \ell(\theta(0); Z) + \phi(Z)^\top z + r(Z, z).$$

Substituting this into (1) gives

$$L_Q(z) - L_Q(0) = \mathbb{E}_P\left[\phi(Z)^\top z\right] + \mathbb{E}_P\left[r(Z, z)\right] + \mathbb{E}_P\left[h(Z)\phi(Z)^\top z\right] + \mathbb{E}_P\left[h(Z)r(Z, z)\right]. \tag{2}$$

Write $\bar{\phi} = \mathbb{E}_P[\phi(Z)]$. Since $\mathbb{E}_P[h] = 0$,

$$\mathbb{E}_P\left[h(Z)\phi(Z)^\top z\right] = \mathbb{E}_P\left[h(Z)(\phi(Z) - \bar{\phi})^\top z\right].$$

Apply Cauchy–Schwarz:

$$\begin{aligned} \left|\mathbb{E}_P\left[h(Z)(\phi(Z) - \bar{\phi})^\top z\right]\right| &\leq \sqrt{\mathbb{E}_P[h(Z)^2]}\sqrt{\mathbb{E}_P\left[\left((\phi(Z) - \bar{\phi})^\top z\right)^2\right]} \\ &\leq \sqrt{\varrho}\sqrt{z^\top \mathbb{E}_P\left[(\phi(Z) - \bar{\phi})(\phi(Z) - \bar{\phi})^\top\right] z} \\ &= \sqrt{\varrho\, z^\top B z}. \end{aligned} \tag{3}$$

Next, use the remainder bound from Assumption 2:

$$|r(Z, z)| \leq \frac{\kappa(Z)}{2}\|z\|^2.$$

Applying Cauchy–Schwarz again,

$$\begin{aligned}
|\mathbb{E}_P\left[h(Z)r(Z,z)\right]| &\le \sqrt{\mathbb{E}_P[h(Z)^2]}\sqrt{\mathbb{E}_P[r(Z,z)^2]} \\
&\le \sqrt{\varrho}\sqrt{\mathbb{E}_P\left[\frac{\kappa(Z)^2}{4}\|z\|^4\right]} \\
&= \frac{\sqrt{\varrho}}{2}\|z\|^2\sqrt{\mathbb{E}_P[\kappa(Z)^2]} \\
&\le \frac{\sqrt{\varrho}\,K}{2}\|z\|^2 . \qquad (4)
\end{aligned}$$

Finally, by Assumption 1,

$$L_P(z) - L_P(0) = a^\top z + \frac{1}{2}z^\top Az.$$

The first two terms on the right-hand side of (2) therefore combine to

$$\mathbb{E}_P\left[\phi(Z)^\top z\right] + \mathbb{E}_P\left[r(Z,z)\right] = L_P(z) - L_P(0) = a^\top z + \frac{1}{2}z^\top Az.$$

Combining this identity with (2), (3), and (4) yields

$$L_Q(z) - L_Q(0) \le a^\top z + \frac{1}{2}z^\top Az + \sqrt{\varrho\, z^\top Bz} + \frac{\sqrt{\varrho}\,K}{2}\|z\|^2 .$$

Adding $L_Q(0)$ to both sides gives the claimed inequality. □

B.2 Proof of Corollary 1

Proof. Start from Theorem 1:

$$L_Q(z) \le L_Q(0) + a^\top z + \frac{1}{2}z^\top Az + \sqrt{\varrho\, z^\top Bz} + \frac{\sqrt{\varrho}\,K}{2}\|z\|^2 .$$

For any $\lambda > 0$ and any $u \ge 0$, Young's inequality gives

$$\sqrt{\varrho\, u} \le \frac{\varrho}{2\lambda} + \frac{\lambda}{2}u.$$

Apply this with $u = z^\top Bz$ to obtain

$$\sqrt{\varrho\, z^\top Bz} \le \frac{\varrho}{2\lambda} + \frac{\lambda}{2}z^\top Bz.$$

Also,

$$\frac{1}{2}z^\top Az + \frac{\sqrt{\varrho}\,K}{2}\|z\|^2 = \frac{1}{2}z^\top\left(A + \sqrt{\varrho}\,KI_r\right)z = \frac{1}{2}z^\top\widetilde{A}z.$$

Substituting both identities into the bound from Theorem 1 yields

$$L_Q(z) \le L_Q(0) + \frac{\varrho}{2\lambda} + a^\top z + \frac{1}{2}z^\top\widetilde{A}z + \frac{\lambda}{2}z^\top Bz.$$

This is the desired claim. □

B.3 Proof of Theorem 2

Proof. Because $A \succ 0$ and $B \succeq 0$, the matrix $A + \lambda B$ is positive definite for every $\lambda > 0$. Hence the quadratic function

$$J_\lambda(z) = a^\top z + \frac{1}{2}z^\top Az + \frac{\lambda}{2}z^\top Bz$$

is strongly convex and has a unique minimizer. Its gradient is

$$\nabla J_\lambda(z) = a + Az + \lambda Bz.$$

Setting the gradient equal to zero yields the normal equation

$$(A + \lambda B)z = -a.$$

Therefore

$$z_\lambda^{\mathrm{unc}} = -(A + \lambda B)^{-1} a.$$

Because $z_{\mathrm{fit}} = -A^{-1}a$, we may rewrite this as

$$z_\lambda^{\mathrm{unc}} = (A + \lambda B)^{-1} A z_{\mathrm{fit}}.$$

This proves the first claim.

For the second claim, because $A \succ 0$ and $B \succeq 0$, the symmetric generalized eigenproblem admits an A-orthonormal basis $V = [v_1, \dots, v_r]$ and nonnegative eigenvalues $\nu_1, \dots, \nu_r$ satisfying

$$BV = AV \operatorname{diag}(\nu_1, \dots, \nu_r), \qquad V^\top AV = I.$$

Expand

$$z_{\mathrm{fit}} = Vc, \qquad c = V^\top A z_{\mathrm{fit}}.$$

Since $z = Vy$ if and only if $y = V^\top Az$, we may write

$$\begin{aligned} J_\lambda(Vy) &= a^\top Vy + \frac{1}{2} y^\top V^\top AVy + \frac{\lambda}{2} y^\top V^\top BVy \\ &= a^\top Vy + \frac{1}{2} y^\top y + \frac{\lambda}{2} y^\top \operatorname{diag}(\nu_1, \dots, \nu_r) y. \end{aligned} \tag{5}$$

Now use $a = -Az_{\mathrm{fit}} = -AVc$, so

$$a^\top Vy = -c^\top V^\top AVy = -c^\top y.$$

Substituting this into (5) gives

$$J_\lambda(Vy) = \sum_{j=1}^{r} \left[-c_j y_j + \frac{1}{2} y_j^2 + \frac{\lambda}{2} \nu_j y_j^2 \right].$$

The coordinates decouple. Minimizing each coordinate separately yields

$$0 = -c_j + (1 + \lambda \nu_j) y_j, \qquad y_j = \frac{c_j}{1 + \lambda \nu_j}.$$

Therefore

$$z_\lambda^{\mathrm{unc}} = Vy = \sum_{j=1}^{r} \frac{c_j}{1 + \lambda \nu_j} v_j.$$

This is exactly the stated generalized eigen-shrinkage formula. □

B.4 Proof of Proposition 1

Proof. The family $\mathcal{F}_{\mathrm{sc}} = \{\alpha z_{\mathrm{fit}} : \alpha \in \mathbb{R}\}$ is a subset of $\mathbb{R}^r$. Since z_λ^{unc} is the global minimizer of J_λ over $\mathbb{R}^r$,

$$\min_{\alpha \in \mathbb{R}} J_\lambda(\alpha z_{\mathrm{fit}}) \geq J_\lambda(z_\lambda^{\mathrm{unc}}).$$

This proves the first inequality.

It remains to characterize equality. By Theorem 2,

$$z_\lambda^{\mathrm{unc}} = \sum_{j=1}^{r} \frac{c_j}{1 + \lambda \nu_j} v_j.$$

Equality holds if and only if $z_\lambda^{\mathrm{unc}} \in \mathcal{F}_{\mathrm{sc}}$, that is, if and only if there exists $\alpha \in \mathbb{R}$ such that

$$\sum_{j=1}^{r} \frac{c_j}{1 + \lambda \nu_j} v_j = \alpha \sum_{j=1}^{r} c_j v_j.$$

Because the generalized eigenvectors form a basis, this is equivalent to

$$\frac{c_j}{1 + \lambda \nu_j} = \alpha c_j \qquad \text{for all } j = 1, \dots, r.$$

For every index j such that $c_j \neq 0$, the condition becomes

$$\alpha = \frac{1}{1+\lambda\nu_j}.$$

Hence the same scalar α can satisfy all active coordinates if and only if

$$\nu_j = \nu_k \qquad \text{for all } j,k \text{ with } c_j \neq 0,\ c_k \neq 0.$$

Conversely, if all active generalized eigenvalues are equal to some common value $\bar{\nu}$, then

$$z_\lambda^{\mathrm{unc}} = \sum_{j=1}^{r} \frac{c_j}{1+\lambda\bar{\nu}} v_j = \frac{1}{1+\lambda\bar{\nu}} z_{\mathrm{fit}},$$

which belongs to $\mathcal{F}_{\mathrm{sc}}$. Therefore equality holds if and only if the generalized eigenvalues are constant on the support of z_{fit}. □

B.5 Proof of Proposition 2

Proof. Because z_λ^{unc} is the minimizer of the quadratic J_λ, the standard completion-of-squares identity around the minimizer gives

$$J_\lambda(z) = J_\lambda(z_\lambda^{\mathrm{unc}}) + \frac{1}{2}(z - z_\lambda^{\mathrm{unc}})^\top (A+\lambda B)(z - z_\lambda^{\mathrm{unc}}) \qquad \text{for all } z.$$

Substituting $z = z_{\mathrm{fit}}$ yields

$$J_\lambda(z_{\mathrm{fit}}) = J_\lambda(z_\lambda^{\mathrm{unc}}) + \frac{1}{2}(z_{\mathrm{fit}} - z_\lambda^{\mathrm{unc}})^\top (A+\lambda B)(z_{\mathrm{fit}} - z_\lambda^{\mathrm{unc}}).$$

Rearranging gives the claimed identity.

The difference is zero if and only if $z_{\mathrm{fit}} = z_\lambda^{\mathrm{unc}}$. From Theorem 2,

$$z_\lambda^{\mathrm{unc}} = (A+\lambda B)^{-1} A z_{\mathrm{fit}}.$$

Hence $z_\lambda^{\mathrm{unc}} = z_{\mathrm{fit}}$ if and only if

$$(A+\lambda B) z_{\mathrm{fit}} = A z_{\mathrm{fit}},$$

which is equivalent to $\lambda B z_{\mathrm{fit}} = 0$, hence $B z_{\mathrm{fit}} = 0$. This proves the second statement. □

B.6 Proof of Theorem 3

Proof. Let

$$M = A + \lambda B, \qquad \hat{M} = \hat{A} + \lambda \hat{B}.$$

By definition,

$$\left\|\hat{M} - M\right\| \leq \left\|\hat{A} - A\right\| + \lambda \left\|\hat{B} - B\right\| = \delta.$$

Since $M \succ 0$ with $\lambda_{\min}(M) = m$ and $\delta < m$, Weyl's inequality implies

$$\lambda_{\min}(\hat{M}) \geq \lambda_{\min}(M) - \left\|\hat{M} - M\right\| \geq m - \delta > 0.$$

Therefore $\hat{M}$ is positive definite, and

$$\left\|\hat{M}^{-1}\right\| \leq \frac{1}{m-\delta}.$$

Now use the exact normal equations

$$M z_\lambda^{\mathrm{unc}} = -a, \qquad \hat{M}\hat{z}_\lambda = -\hat{a}.$$

Subtract the first from the second:

$$\hat{M}(\hat{z}_\lambda - z_\lambda^{\mathrm{unc}}) = -(\hat{a} - a) - (\hat{M} - M) z_\lambda^{\mathrm{unc}}.$$

Multiply by $\hat{M}^{-1}$:

$$\hat{z}_\lambda - z_\lambda^{\mathrm{unc}} = -\hat{M}^{-1}(\hat{a} - a) - \hat{M}^{-1}(\hat{M} - M)z_\lambda^{\mathrm{unc}}.$$

Taking norms and using the bound on $\hat{M}^{-1}$,

$$\begin{aligned}\|\hat{z}_\lambda - z_\lambda^{\mathrm{unc}}\| &\le \left\|\hat{M}^{-1}\right\| \left[\|\hat{a} - a\| + \left\|\hat{M} - M\right\| \|z_\lambda^{\mathrm{unc}}\|\right] \\ &\le \frac{\|\hat{a} - a\| + \delta \|z_\lambda^{\mathrm{unc}}\|}{m - \delta}.\end{aligned}$$

This proves the deterministic perturbation inequality.

If $\hat{a} \to a$, $\hat{A} \to A$, and $\hat{B} \to B$, then $\delta \to 0$ and $\|\hat{a} - a\| \to 0$. The right-hand side therefore converges to zero, implying $\hat{z}_\lambda \to z_\lambda^{\mathrm{unc}}$. □

B.7 Proof of Corollary 2

Proof. Corollary 1 states that for every $z \in \mathcal{Z}$,

$$L_Q(z) \le L_Q(0) + \frac{\varrho}{2\lambda} + J_\lambda(z).$$

Apply this inequality at $z = z_\lambda^{\mathcal{Z}}$:

$$L_Q(z_\lambda^{\mathcal{Z}}) \le L_Q(0) + \frac{\varrho}{2\lambda} + J_\lambda(z_\lambda^{\mathcal{Z}}).$$

Because $z_\lambda^{\mathcal{Z}}$ minimizes J_λ over $\mathcal{Z}$,

$$J_\lambda(z_\lambda^{\mathcal{Z}}) \le J_\lambda(z) \qquad \text{for all } z \in \mathcal{Z}.$$

Combining the two inequalities yields

$$L_Q(z_\lambda^{\mathcal{Z}}) \le L_Q(0) + \frac{\varrho}{2\lambda} + J_\lambda(z_\lambda^{\mathcal{Z}}) \le L_Q(0) + \frac{\varrho}{2\lambda} + J_\lambda(z) \qquad \text{for all } z \in \mathcal{Z}.$$

This proves the corollary. □

C Optional Numerical Properties

Although SWING has a closed-form solution, one useful limit statement is worth recording.

Proposition 3 (Shrinkage path limits)**.** *Under the assumptions of Theorem 2, let*

$$z_\lambda^{\mathrm{unc}} = \sum_{j=1}^{r} \frac{c_j}{1 + \lambda\nu_j} v_j.$$

Then:

1. $z_\lambda^{\mathrm{unc}} \to z_{\mathrm{fit}}$ *as* $\lambda \downarrow 0$*;*
2. $z_\lambda^{\mathrm{unc}} \to \sum_{\nu_j = 0} c_j v_j$ *as* $\lambda \to \infty$*;*
3. *for every* j *with* $\nu_j > 0$*, the coefficient* $\frac{c_j}{1+\lambda\nu_j}$ *is continuous in* λ *and strictly decreases in magnitude as* λ *increases.*

Proof. All three claims are immediate from the closed-form coefficient

$$\frac{c_j}{1 + \lambda\nu_j}.$$

If $\lambda \downarrow 0$, then $1 + \lambda\nu_j \to 1$, so each coefficient converges to c_j. If $\lambda \to \infty$, then the coefficient converges to 0 for every $\nu_j > 0$ and remains c_j for every $\nu_j = 0$. The continuity and monotonicity follow from elementary calculus because $\nu_j \ge 0$. □